

2017 HKDSE Physics

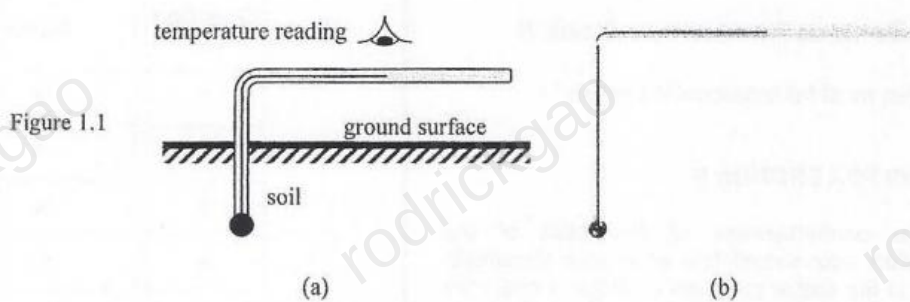
Paper 1B — Structured Questions

Original questions and official answers with detailed English reasoning. Calculations show relationships, substitution and units; each structured subpart is presented separately.

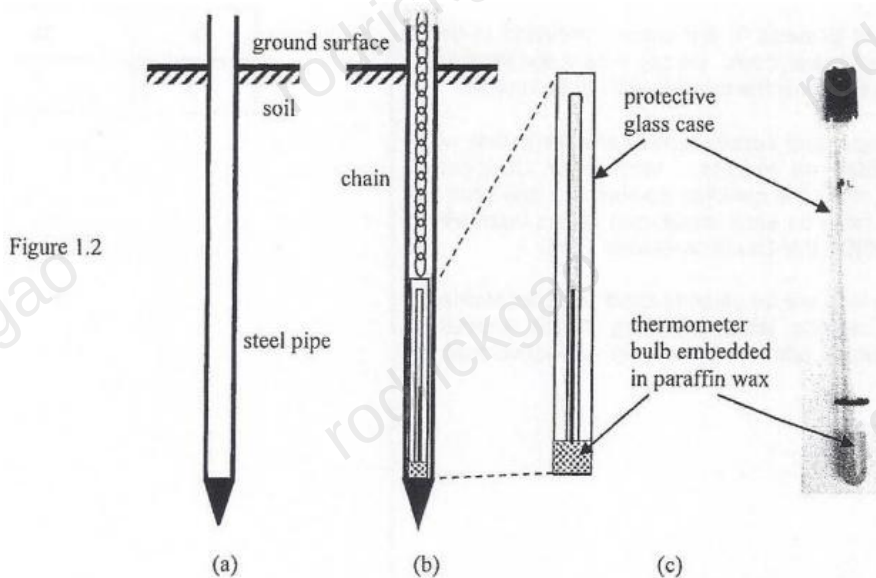
Question 1

Read the following passage about soil thermometer and answer the questions that follow.

The temperature of soil changes with depth, and this information is important to farmers and scientists. To measure soil temperatures at depths close to the ground surface, the bulb of a thermometer is buried in the soil. The stem of the thermometer is bent 90° for easy reading. Figure 1.1a is a schematic diagram and Figure 1.1b shows a photo of a soil thermometer.



For depths greater than 30 cm, a steel pipe is driven into the soil (Figure 1.2a); and a liquid-in-glass thermometer with a protective glass case is lowered into the steel pipe (Figure 1.2b). The bulb of the thermometer is embedded in paraffin wax (Figure 1.2c). To read the temperature, the thermometer is lifted out of the steel pipe by pulling the chain.



(a) As shown in Figure 1.1b, the bulb of the soil thermometer is very large compared to those of common thermometers. Suggest a reason for this design. (1 mark)

(b) On a certain morning, the air temperature is 15°C . An observer takes a measurement of the soil temperature at 1 m deep. The thermometer reading is 20°C . It is given that the mass of the paraffin wax enclosing the thermometer bulb is 0.015 kg , and the specific heat capacity of paraffin wax is $2.9 \times 10^3\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$.

- (i) Calculate the energy loss of the paraffin wax as it cools down to the air temperature. (2 marks)
- (ii) It is known that the paraffin wax enclosing the bulb of the thermometer gains or loses energy at a constant rate of 0.5 J s^{-1} , estimate the time taken for the paraffin wax to reach the air temperature after the thermometer is lifted out of the soil. (2 marks)
- (iii) If there is no paraffin wax enclosing the bulb of the thermometer, explain how the thermometer reading as recorded by the observer is affected. (2 marks)

Official marking scheme

Solution		Marks	Remarks
1.	(a) A larger bulb improves the sensitivity of the thermometer.	1A	
	Or A larger bulb minimizes the effect on the temperature reading due to the other parts of the thermometer stem that are exposed to different temperatures.	1A	
		1	
	(b) (i) $E = mc\Delta T$ $= 0.015 \times (2.9 \times 10^3) \times (20 - 15)$ $= 217.5\text{ J}$	1M 1A	
		2	
	(ii) Time taken to reach air temperature $= \frac{217.5}{0.5}$ $= 435\text{ s}$	1M 1A	
		2	
	(iii) The thermometer would be in direct contact with the cooler air and would cool down quickly.	1A	
	The temperature reading would be less than the actual soil temperature.	1A	
		2	

(a) A larger bulb contains more thermometric liquid. For the same temperature rise, its larger volume expansion causes a larger rise in a capillary of the same bore, increasing sensitivity.

(b)(i) Heat lost is $Q = mc\Delta T = 0.015(2.9 \times 10^3)(20 - 15) = 217.5\text{ J}$.

(b)(ii) At the stipulated constant heat-loss rate, $t = Q/P = 217.5/0.5 = 435\text{ s}$, about 7.25 min.

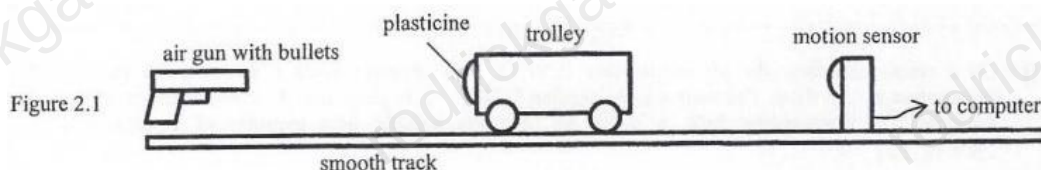
(b)(iii) Without the wax, the bulb has less thermal buffering and cools more quickly after removal into the colder air. The reading recorded would therefore be below the actual soil temperature of 20°C .

Question 2

The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track; a trolley; a motion sensor used to measure the speed of the trolley; some plasticine; an air gun and bullets; an electronic balance

The set-up is shown in Figure 2.1.



Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down ONE precaution for getting a more accurate result. (5 marks)

Official marking scheme

2. Measure the mass of a bullet m and the mass of the trolley with plasticine M .	1A
Fire the bullet towards the plasticine.	1A
Read the speed of the trolley v immediately after the bullet hit the plasticine.	1A
The speed of the bullet u is given by $u = \frac{M+m}{m}v$.	1A
Precaution:	1A
- The bullet should be fired close to the plasticine.	Any ONE
- The bullet should be fired along the direction of travel of the trolley.	
- The track must be horizontal / friction compensated.	
	5

Measurement: Use the balance to measure bullet mass m and the mass M of the trolley together with its plasticine. Place the trolley initially at rest on the track.

Procedure: Fire the bullet horizontally into the plasticine so that it remains embedded. Use the motion sensor to measure the common speed v immediately after impact.

Principle and equation: Neglect external horizontal impulse during the short collision. Conservation of momentum gives $mu = (M+m)v$, hence the bullet's initial speed $u = (M+m)v/m$. Kinetic energy is not conserved in this embedding collision.

Precaution: Align the shot with the trolley's direction of travel. This avoids sideways momentum and unwanted rotation. A level, effectively frictionless track is also necessary for the stated method.

Question 3

The average kinetic energy of one monatomic gas molecule at temperature T is given by

$$E_k = (3/2)(R/N_A)T,$$

where R is the universal gas constant and N_A is the Avogadro constant. A monatomic gas is heated from 300 K to 350 K under fixed volume.

(a) Estimate the ratio of the root-mean-square speed (c_{rms}) of the gas molecules at the two temperatures:

c_{rms} at 350 K / c_{rms} at 300 K. (2 marks)

(b) Thus, using kinetic theory, explain why the gas pressure would increase. (2 marks)

Official marking scheme

Solution	Marks	Remarks
3. (a) $\frac{(c_{\text{rms}})_f}{c_{\text{rms}}} = \sqrt{\frac{T_f}{T}}$	1M	
$\frac{c_{\text{r.m.s. at 350 K}}}{c_{\text{r.m.s. at 300 K}}} = \sqrt{\frac{350}{300}}$	1A	
$= 1.08$	2	
(b) The speed of the gas molecules increases. They collide more frequently and violently with the wall of the container. Thus, the pressure increases.	1A+1A	
	2	

(a) Since $\frac{1}{2}mc_{\text{rms}}^2 = (3/2)(R/N_A)T$, c_{rms} is proportional to $\sqrt{T}$. The required ratio is $\sqrt{(350/300)} = 1.08$. The temperatures are already in kelvin.

(b) Faster molecules strike the walls more frequently and transfer more momentum per collision on average. The rate of momentum transfer to the walls, hence force per unit area and pressure, increases at fixed volume.

Question 4

(a) A steel ball bearing is released from rest at time $t = 0$. A stroboscopic photo is taken at 0.05 s time intervals. The results are shown in Figure 4.1. Neglect air resistance.

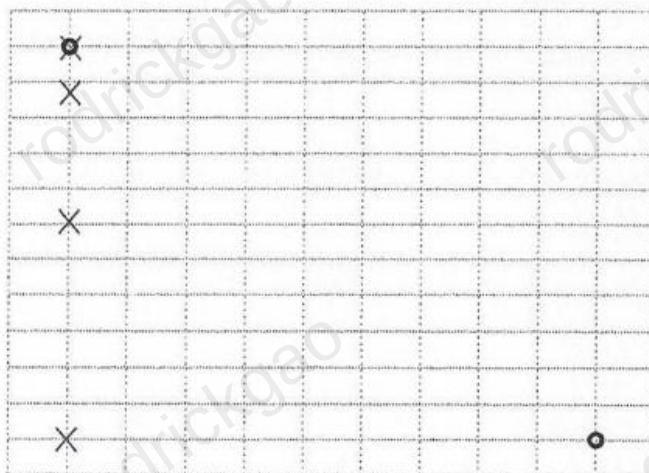
Figure 4.1



(i) Estimate the acceleration due to gravity using the data in Figure 4.1. (2 marks)

(ii) The bearing is now projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles ($\circ$) in Figure 4.2. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using crosses ($\times$).

Figure 4.2

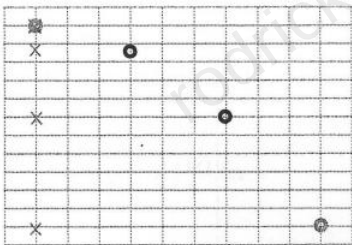


(a)(ii)(1) In Figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles ($\circ$). (2 marks)

(a)(ii)(2) Given that the bearing is projected horizontally with an initial speed of 1 m s^{-1} , use the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken. (3 marks)

(b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's laws of motion, explain why the speed of the ball becomes constant. (3 marks)

Official marking scheme

4. (a) (i)	By $s = ut + \frac{1}{2}gt^2$ $0.11 = \frac{1}{2}g(0.05 \times 3)^2$ $g = 9.78 \text{ m s}^{-2}$	1M 1A	2
(ii) (1)		1A 1A	2
(2)	$v_x = 1 \text{ m s}^{-1}$ $v_y = u_y + gt$ $= 0 + 9.78 \times (0.05 \times 3)$ $= 1.47 \text{ m s}^{-1}$ $v = \sqrt{v_x^2 + v_y^2}$ $= \sqrt{1^2 + 1.47^2}$ $= 1.78 \text{ m s}^{-1}$	1M 1M 1A	3
(b)	The air resistance acting on the ball increases as its speed increases. When the air resistance equals to the weight of the ball, net force acting on the ball becomes zero, by Newton's first law of motion, the ball travels with constant speed. <u>Or</u> net force acting on the ball becomes zero, by Newton's second law of motion, the ball will not accelerate further and travels with constant speed.	1A 1A 1A 1A	3

(a)(i) The last image is three 0.05 s intervals after release, so $t = 0.15 \text{ s}$. From the graph, vertical displacement is 0.11 m. With $u = 0$, $s = \frac{1}{2}gt^2$ gives $g = 2(0.11)/(0.15)^2 = 9.78 \text{ m s}^{-2}$ downward.

(a)(ii)(1) Put the two intermediate circles at equally spaced horizontal positions between the first and last circles. Their vertical coordinates must equal those of the corresponding crosses, because horizontal projection does not change the vertical free-fall motion.

(a)(ii)(2) Horizontal velocity remains $v_x = 1.00 \text{ m s}^{-1}$. At $t = 0.15 \text{ s}$, $v_y = gt = 9.78(0.15) = 1.47 \text{ m s}^{-1}$ downward. Speed $v = \sqrt{v_x^2 + v_y^2} = 1.78 \text{ m s}^{-1}$.

(b) Weight acts downward and air resistance upward. As speed increases, air resistance increases until it balances weight. The net force and acceleration are then zero, so the ball continues at constant terminal speed.

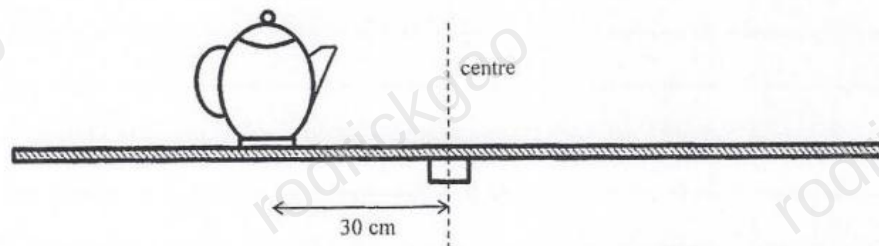
Question 5

The photo shows a turntable commonly used in restaurants.



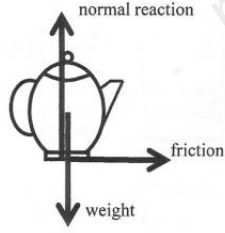
A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable. Figure 5.1 shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

Figure 5.1



- (a) On Figure 5.1, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)
- (b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)
- (c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

Official marking scheme

Solution	Marks	Remarks
5. (a) 	1A+1A	
	2	
(b) $\omega = \pi \text{ s}^{-1}$ $F = mr\omega^2$ $= (1)(0.3)(\pi)^2$ $= 2.96 \text{ N (towards the centre of the turntable)}$	1A 1M 1A	
<i>Alternative method:</i> $v = 0.3\pi \text{ m s}^{-1}$ $F = m \frac{v^2}{r}$ $= 2.96 \text{ N}$	1A 1M 1A	
	3	
(c) The initial linear speed of the teapot $= r\omega = 0.3\pi \text{ m s}^{-1}$ Deceleration of the teapot $a = \frac{f}{m} = \frac{10}{1} = 10 \text{ m s}^{-2}$ Distance travelled s is given by $v^2 - u^2 = 2as$ $s = \frac{u^2}{2a} = \frac{(0.3\pi)^2}{2(10)}$ $= 0.044 \text{ m (or 4.4 cm)}$	1M 1M 1A	
<i>Alternative method:</i> The initial linear speed of the teapot $= r\omega = 0.3\pi \text{ m s}^{-1}$ K.E. of the teapot is dissipated as work done against friction $\frac{1}{2} mu^2 = fd$ $d = \frac{mu^2}{2f} = \frac{(1)(0.3\pi)^2}{2(10)}$ $= 0.044 \text{ m}$	1M 1M 1A	
	3	

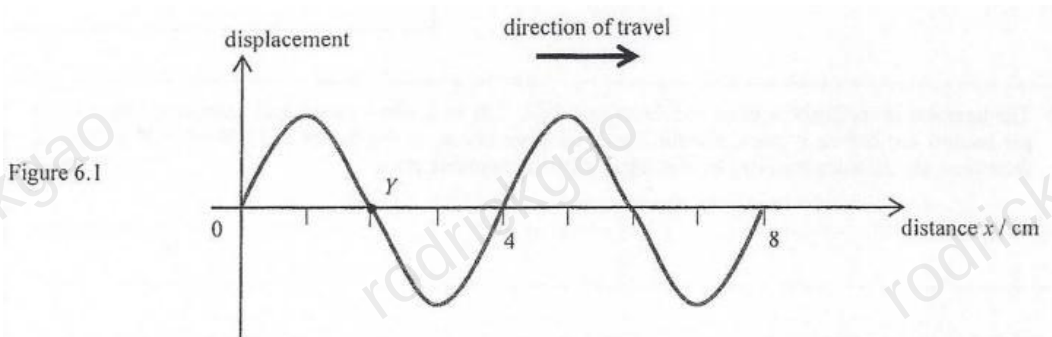
(a) Draw weight mg vertically downward, normal reaction N vertically upward, and static friction horizontally towards the centre of the turntable. The vertical forces balance; friction supplies the net centripetal force, not an additional fourth force.

(b) Angular speed $\omega = 2\pi f = 2\pi(0.5) = \pi \text{ rad s}^{-1}$. Net force $F = mr\omega^2 = 1(0.30)\pi^2 = 2.96 \text{ N}$ towards the centre.

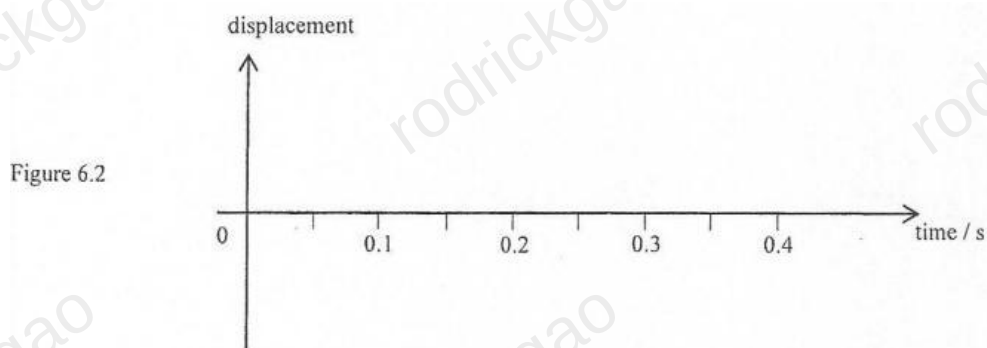
(c) Immediately before stopping, the teapot's speed is $u = r\omega = 0.30\pi = 0.942 \text{ m s}^{-1}$. Work against the constant 10 N friction removes its kinetic energy: $10s = \frac{1}{2}(1)(0.942)^2$. Thus $s = 0.0444 \text{ m} \approx 4.4 \text{ cm}$, assuming it remains on the turntable.

Question 6

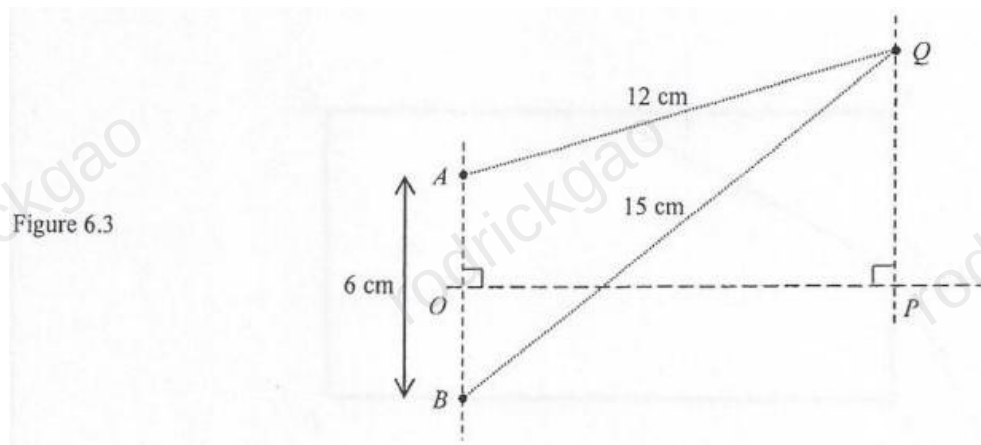
(a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure 6.1 shows the displacement-distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.



- Determine the wave speed of the water wave. (2 marks)
- State the direction of motion of particle Y at $t = 0$. (1 mark)
- Sketch the displacement-time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure 6.2. (2 marks)

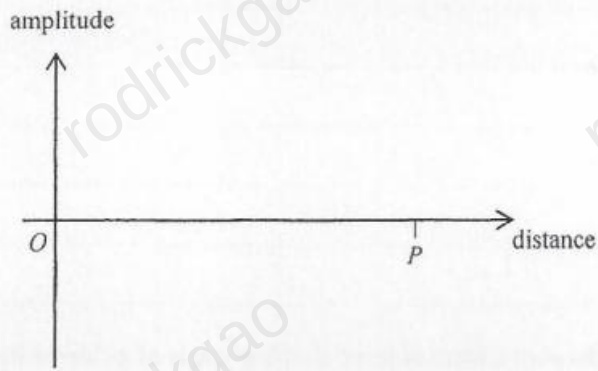


(b) In Figure 6.3, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB. Q is a second minimum from P, where $AQ = 12$ cm and $BQ = 15$ cm.



- Explain why a minimum occurs at Q. (2 marks)
- Determine the wavelength of the water wave. (2 marks)
- Sketch in Figure 6.4 how the AMPLITUDE of the water wave varies along the line OP. (1 mark)

Figure 6.4



Official marking scheme

Solution		Marks	Remarks
6. (a)	(i) $v = f\lambda$ $= 5 \times 4$ $= 20 \text{ cm s}^{-1}$	1M 1A	
		2	
	(ii) Y is moving upwards at $t = 0$.	1A	
		1	
	(iii)		
		1A 1M	
		2	
	(b) (i) The water waves from A and B are in anti-phase at Q. <u>Or</u> The path difference at Q = $(n + 1/2)\lambda$. Destructive interference occurs to form a minimum.	1M 1M 1A	
		2	
	(ii) Path difference at Q = $1.5\lambda = 3 \text{ cm}$ $\lambda = 2 \text{ cm}$	1M 1A	
		2	
	(iii)		
		1A	
		1	

(a)(i) The graph gives $\lambda = 4 \text{ cm} = 0.040 \text{ m}$. Hence $v = f\lambda = 5(0.040) = 0.20 \text{ m s}^{-1}$, or 20 cm s^{-1} .

(a)(ii) Y moves upward at $t = 0$: the right-travelling waveform carries the neighbouring positive displacement into Y.

(a)(iii) $T = 1/f = 0.20 \text{ s}$. Draw two complete sinusoidal cycles from 0 to 0.40 s, starting at zero displacement with positive slope. Maxima occur at 0.05 and 0.25 s; minima at 0.15 and 0.35 s.

(b)(i) The two in-phase sources produce waves arriving in antiphase at Q. Their displacements cancel by destructive interference, forming a minimum.

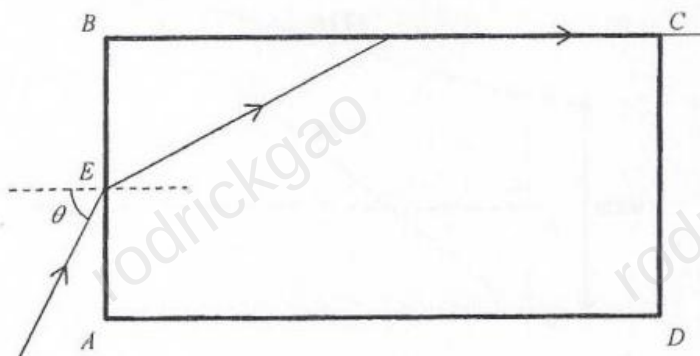
(b)(ii) The second minimum corresponds to path difference $3\lambda/2$. Thus $15 - 12 = 3\lambda/2$ in centimetres, giving $\lambda = 2.0 \text{ cm}$.

(b)(iii) Along OP, the two path lengths are equal, so interference remains constructive. Draw a positive amplitude that decreases smoothly from O towards P as the waves spread out; do not draw alternating maxima and minima.

Question 7

(a) A light ray enters a rectangular plastic block ABCD from air at point E, and the angle of incidence is θ . The light ray emerges along face BC as shown in Figure 7.1. The refractive index of the plastic is 1.36.

Figure 7.1



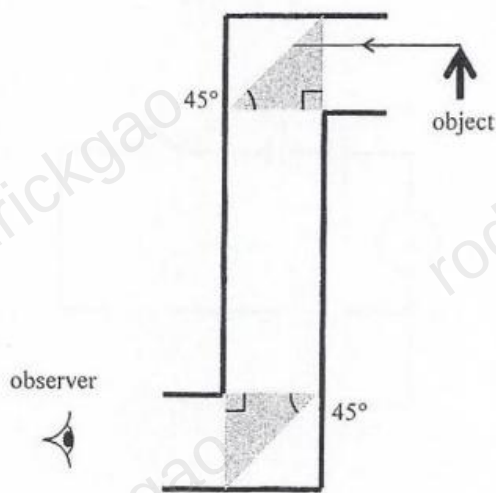
(i) Find the critical angle of the plastic. (2 marks)

(ii) Find the value of θ . (3 marks)

(iii) If the light ray enters the plastic block at point E with an angle of incidence larger than θ , sketch the path of the light ray in Figure 7.1. (2 marks)

(b) A student designs a periscope using two plastic prisms; the refractive index of the plastic is 1.36. As shown in Figure 7.2, an object is placed in front of the periscope.

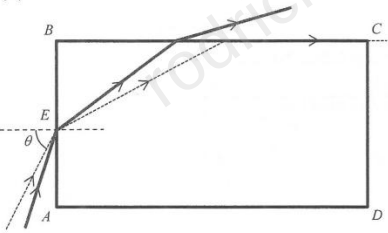
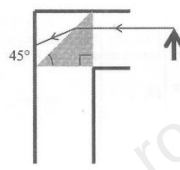
Figure 7.2



(i) Complete the path of the light ray from the object in Figure 7.2, and explain why the periscope fails to work. (3 marks)

(ii) What can be used to replace the two plastic prisms so that the periscope can work properly? (1 mark)

Official marking scheme

Solution		Marks	Remarks
7. (a)	(i) At the critical angle c , $\frac{\sin 90^\circ}{\sin c} = n$ $\frac{1}{\sin c} = 1.36$ $c = 47.3^\circ$	1M 1A	
		2	
	(ii) Angle of refraction at $E = 90^\circ - 47.33^\circ = 42.67^\circ$ By Snell's law $\frac{\sin \theta}{\sin 42.67^\circ} = 1.36$ $\theta = 67.2^\circ$	1M 1M 1A	
		3	
	(iii) 	2A	
		2	
	(b) (i) 	1A	
	The angle of incidence of the light ray from the object is (45° , which is) less than the critical angle of the plastic prism. Total internal reflection will not occur and the image may not be clear enough for observation.	1A 1A 1A	
		3	
	(ii) Glass prism (with critical angle less than 45°) Or Plane mirror	1A 1A	
		1	

(a)(i) At the plastic–air boundary, $\sin c = 1/n = 1/1.36$. Thus $c = 47.3^\circ$.

(a)(ii) The two face normals are perpendicular, so the refracted angle at E is $r = 90^\circ - c = 42.7^\circ$. Snell's law gives $\sin \theta = 1.36 \sin r$, hence $\theta = 67.2^\circ$.

(a)(iii) Increasing θ increases r at E and decreases the angle of incidence at BC below c . Draw the ray refracting out through BC, away from its normal, rather than travelling along BC or undergoing total internal reflection.

(b)(i) The ray enters the first prism normally, then strikes its diagonal face at 45° . Since $45^\circ < 47.3^\circ$, total internal reflection does not occur. Draw a transmitted ray refracting out through that face; insufficient light follows the required periscope path.

(b)(ii) Replace the prisms by plane mirrors at 45° , or use prisms with refractive index greater than $\sqrt{2}$ so their critical angle is below 45° .

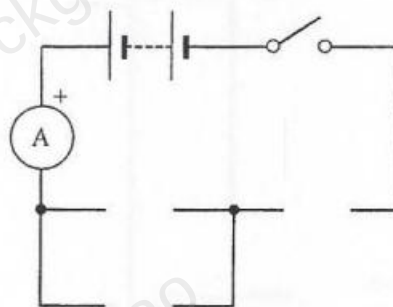
Question 8

A student uses the following apparatus to measure the resistance of a tungsten filament light bulb.

a battery; a switch; a variable resistor; an ammeter; a voltmeter; a light bulb

(a) Figure 8.1 shows an incomplete circuit for the experiment. The “+” symbol represents the positive terminal of the ammeter. Use suitable circuit symbols to complete the circuit, and mark the positive terminal of the voltmeter with “+”. (3 marks)

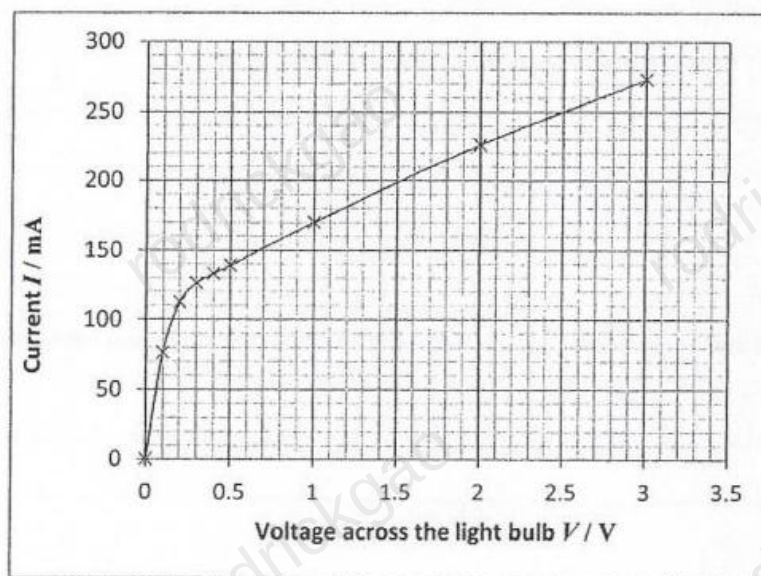
Figure 8.1



The table below and Figure 8.2 show the results obtained.

V / V	0	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0
I / mA	0	76	112	126	133	139	170	226	273

Figure 8.2



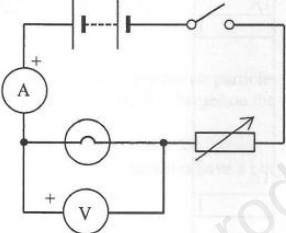
(b) Briefly explain the variation of the resistance of the light bulb with the voltage across the light bulb. (2 marks)

(c) The student claims that since the resistance of the light bulb is not a constant, the equation $R = V/I$ cannot be used to calculate the resistance of the light bulb. Briefly explain why his claim is wrong. (1 mark)

(d) Determine the resistance of the light bulb at $V = 0.1$ V and 2.5 V. (3 marks)

(e) It is given that the cross-sectional area of the tungsten filament in the light bulb is $1.66 \times 10^{-9} \text{ m}^2$, and the resistivity of tungsten at room temperature is about $5.6 \times 10^{-8} \Omega \text{ m}$. Estimate the length of the tungsten filament in the light bulb using the appropriate resistance found in (d). (3 marks)

Official marking scheme

Solution	Marks	Remarks
8. (a) 	1A 1A 1A	Correct symbols of light bulb, variable resistor and voltmeter Correct positions Correct positive terminal connection for the voltmeter
	3	
(b) As the voltage across the light bulb increases, the temperature of the light bulb increases, thus its resistance increases.	1A 1A	
	2	
(c) $R = \frac{V}{I}$ is the definition of resistance. It is applicable to all conductors.	1A	
	1	
(d) At $V = 0.1 \text{ V}$ $R = \frac{V}{I} = \frac{0.1}{76 \times 10^{-3}} = 1.32 \Omega$ At $V = 2.5 \text{ V}$ $R = \frac{V}{I} = \frac{2.5}{250 \times 10^{-3}} = 10 \Omega$	1A 1A 1A	for reading correct values (ignore the order of magnitude) from the graph
	3	
(e) $R = \rho \frac{l}{A}$ $l = \frac{RA}{\rho}$ $= \frac{1.32 \times (1.66 \times 10^{-9})}{5.6 \times 10^{-8}}$ $= 0.039 \text{ m}$	1M+1M 1A	
	3	

(a) Put the bulb and variable resistor in series with the battery, switch and ammeter. Connect the voltmeter in parallel across the bulb, with its positive terminal on the side connected towards the battery's positive terminal (the left side in the given circuit).

(b) Increasing the voltage heats the tungsten filament. Its resistivity rises with temperature, so the resistance increases.

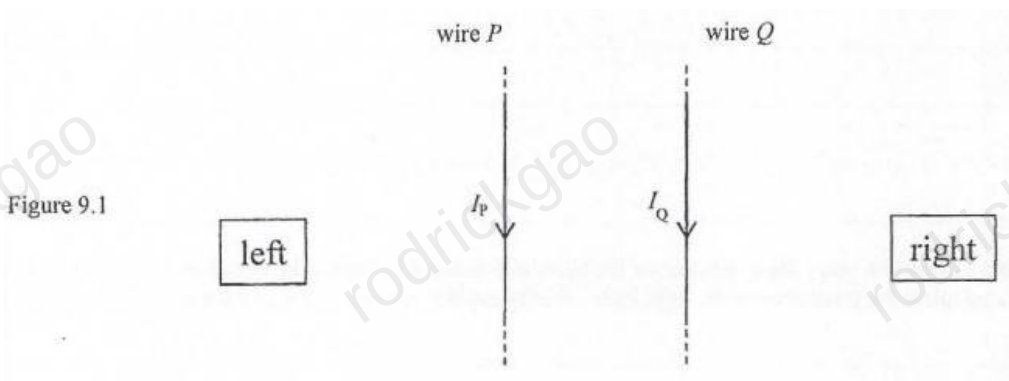
(c) $R = V/I$ defines resistance at each operating point, even for a non-ohmic component. Ohm's law requires a constant ratio under unchanged physical conditions; the definition does not require a constant resistance.

(d) At 0.10 V , $I = 76 \text{ mA}$, so $R = 0.10/(76 \times 10^{-3}) = 1.32 \Omega$. At 2.5 V , the graph gives $I \approx 250 \text{ mA}$, so $R \approx 2.5/0.250 = 10 \Omega$. Use V/I , not the tangent slope of the I - V curve.

(e) Choose the low-voltage resistance, because the given resistivity is at room temperature. From $R = \rho l/A$, $l = RA/\rho = 1.32(1.66 \times 10^{-9})/(5.6 \times 10^{-8}) \approx 0.039 \text{ m} = 3.9 \text{ cm}$. This estimates the filament as nearly at room temperature at 0.10 V .

Question 9

(a) Two long straight current-carrying wires, P and Q, are parallel to each other and lie on the plane of the paper as shown in Figure 9.1. The currents in the wires, I_P and I_Q , flow in the same direction.



(i) State the direction (to the left / to the right / into the paper / out of the paper) of the magnetic field at Q due to P. (1 mark)

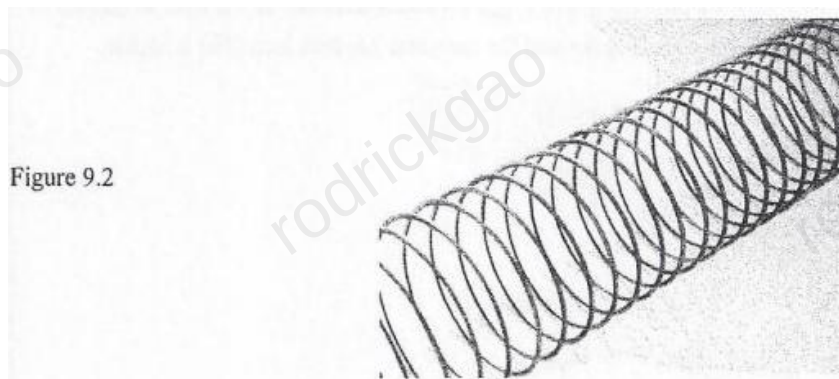
(ii) In Figure 9.1, draw the direction of the magnetic force acting on Q due to P. (1 mark)

(iii) Show that the magnitude of the magnetic force per unit length F_1 acting on Q due to P is

$$F_1 = \mu_0 I_P I_Q / (2\pi r),$$

where μ_0 is the permeability of free space and r is the separation between the two wires. (3 marks)

(iv) For the magnetic force acting on Q due to P and the magnetic force acting on P due to Q, if $I_P \neq I_Q$, briefly explain whether the two forces are equal in magnitude. (2 marks)

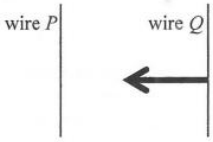


(b) Figure 9.2 shows a metal slinky spring.

(i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

(ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

Official marking scheme

Solution			Marks	Remarks
9.	(a)	(i) The magnetic field at Q due to P points out of the paper.	1A	
			1	
	(ii)		1A	
			1	
	(iii)	The magnetic field at Q due to P is $B_Q = \frac{\mu_0 I_P}{2\pi r}$ For a certain length segment l of wire, the magnetic force is $F = B_Q I_Q l \sin \theta$ $= \frac{\mu_0 I_P}{2\pi r} I_Q l$ The magnetic force per unit length is $F_l = \frac{F}{l} = \frac{\mu_0 I_P I_Q}{2\pi r}$	1M	
			1M	
			1M	
			3	
	(iv)	The two forces form an action and reaction pair, thus they are equal in magnitude.	1A	
			1A	
			2	
	(b)	(i) As current passes in the same direction between neighbouring wire segments, the wire segments attract each other, and the solenoid is compressed.	1A	
			1A	
			2	
		(ii) Current keeps on flowing in the same direction between neighbouring wire segments at each instant, thus the solenoid will be compressed due to magnetic force.	1A	
			1	

(a)(i) The magnetic field at Q due to P is out of the paper, from the right-hand grip rule for the indicated current.

(a)(ii) Draw the force on Q to the left, towards P . Parallel currents in the same direction attract.

(a)(iii) At separation r , $B_Q = \mu_0 I_P / (2\pi r)$. For a length l of Q perpendicular to this field, $F = B_Q I_Q l$. Dividing by l gives $F_l = F/l = \mu_0 I_P I_Q / (2\pi r)$.

(a)(iv) The forces are equal in magnitude and opposite in direction by Newton's third law, even when $I_P \neq I_Q$. The expression contains the same product $I_P I_Q$ for either wire.

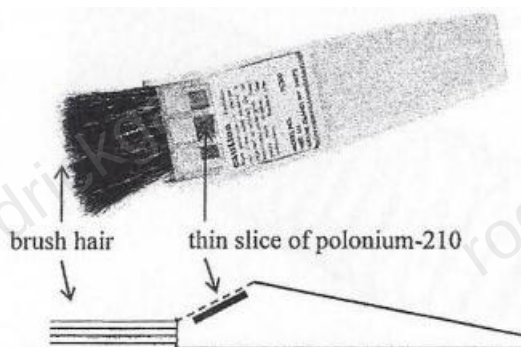
(b)(i) Neighbouring turns carry current in the same direction and attract, so the spring is compressed.

(b)(ii) With alternating current, neighbouring turns reverse current together and still attract at each instant. The force magnitude varies, approximately as I^2 , but its direction does not alternate between attraction and repulsion.

Question 10

Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

Figure 10.1



- Write a nuclear equation for the decay of polonium-210. (2 marks)
- Briefly explain how the α particles help clean the charged dust. (2 marks)
- Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)
- The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days).
Given: half-life of polonium-210 is 138 days. (2 marks)

Official marking scheme

Solution	Marks	Remarks
10. (a) $^{210}_{84}\text{Po} \rightarrow ^{206}_{82}\text{Pb} + ^4_2\text{He}$	2A	
	2	
(b) The α -particles ionize the air particles. The ions neutralize the charges on the dust/photo or film surface.	1A 1A	
	2	
(c) This is because α -particles have a range of only a few centimeters in air.	1A	
	1	
(d) Activity after 1 year = $\left(\frac{1}{2}\right)^{\frac{365}{138}}$ = 0.160 unit	1M 1A	
<i>Alternative method:</i> $A = A_0 e^{-\frac{\ln 2}{t_{1/2}} t}$ $= 1 \times e^{-\frac{\ln 2}{138}(365)}$ $= 0.160 \text{ unit}$	1M 1A	
	2	

- (a) ${}_{84}^{210}\text{Po} \rightarrow {}_{82}^{206}\text{Pb} + {}_2^4\text{He}$. Both nucleon number ($210 = 206 + 4$) and proton number ($84 = 82 + 2$) are conserved.
- (b) Alpha particles ionize nearby air molecules. Ions of the appropriate sign neutralize charges on the dust and the photo or film, reducing electrostatic attraction so the dust can be brushed away.
- (c) Alpha particles have a range of only a few centimetres in air, so the source must be close to the brush hair and dust.
- (d) $A/A_0 = (1/2)^{t/T} = (1/2)^{365/138} = 0.160$. With $A_0 = 1$ unit, activity after one year is 0.160 unit, about 16% of its initial value.